

Unit 8, Lesson 6: Completing the Square when $a > 1$ Using Algebra Tiles



Benchmarks

MA.912.AR.3.6 Given an expression or equation representing a quadratic function, determine the vertex and zeros, and interpret them in terms of a real-world context.

MA.912.AR.1.2 Rearrange equations or formulas to isolate a quantity of interest.



Learning Target

- ☐ I can use algebra tiles to complete the square for a quadratic expression where $a > 1$.



8.6.1 Warm-Up: Bell Work

1. Use the diagram to complete the square for the expression, $x^2 + 10x + 8$.

	x	1	1	1	1	1
x	x^2	x	x	x	x	x
1	x	1	1	1	1	1
1	x	1	1	1		
1	x					
1	x					
1	x					



8.6.2 Exploration: Completing the Square Where $a > 1$ Using Algebra Tiles

1. The diagram shows three perfect square trinomials.

	x	1	1	1
x	x^2	x	x	x
1	x	1	1	1
1	x	1	1	1
1	x	1	1	1

	x	1	1	1
x	x^2	x	x	x
1	x	1	1	1
1	x	1	1	1
1	x	1	1	1

	x	1	1	1
x	x^2	x	x	x
1	x	1	1	1
1	x	1	1	1
1	x	1	1	1

Complete each representation of the diagram.

Three Perfect Square Trinomials	Three Binomials Squared	All Like Terms Added
$3(x^2 + \underline{\hspace{1cm}}x + \underline{\hspace{1cm}})$	$3(x + \underline{\hspace{1cm}})^2$	$\underline{\hspace{1cm}}x^2 + \underline{\hspace{1cm}}x + \underline{\hspace{1cm}}$

2. The expression $3(x^2 + 6x) + 11$ is represented in the diagram using three product mats.

	x	1	1	1
x	x^2	x	x	x
1	x	1	1	1
1	x	1	1	1
1	x	1	1	1

	x	1	1	1
x	x^2	x	x	x
1	x	1	1	
1	x			
1	x			

	x	1	1	1
x	x^2	x	x	x
1	x			
1	x			
1	x			

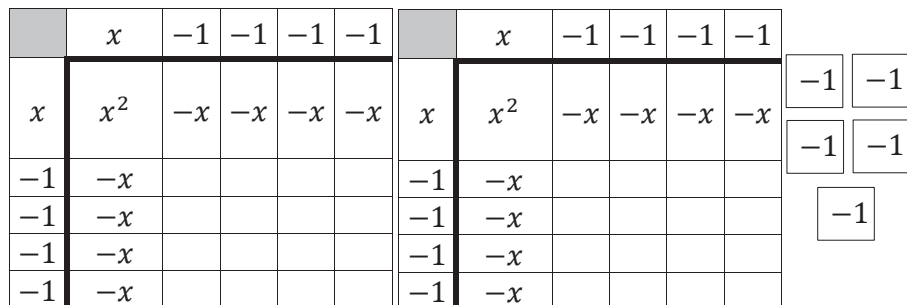
- Discuss this representation with your partner.
- Name the property of operations that is applied to rewrite $3x^2 + 18x + 11$ as $3(x^2 + 6x) + 11$.
- How many unit tiles (1) should be added to complete all of the squares?

By adding those unit tiles, the diagram now shows $3(x + 3)^2$. As a reminder, if 1-unit tiles are added to the model, then the same number of -1 -unit tiles must be added to the model so that its value doesn't change. Algebraically, this can be represented by subtracting that number of units.

- Complete the expression.

$$3(x + 3)^2 - \underline{\hspace{2cm}}$$

3. The diagram of algebra tiles shown represents the expression $2x^2 - 16x - 5$.



- a. Rewrite the expression $2x^2 - 16x - 5$ to show the representation of the two separate squares by factoring 2 out from the first two terms.

$$2(x^2 \underline{\hspace{1cm}}) - 5$$

- b. How many unit tiles (1) should be added to complete **one** of the squares?
- c. Add the number of unit tiles for **one** square to the expression inside the parentheses.

$$2(x^2 - 8x + \underline{\hspace{1cm}}) - 5$$

- d. Describe how the coefficient of x is related to the number of unit tiles added for **one** square.

$$2(x^2 - 8x + \underline{\hspace{1cm}}) - 5$$


- e. Discuss with your partner how the 2 outside of the parentheses is related to the diagram.
- f. What is the total number of unit tiles added to complete the squares?
- g. Discuss with your partner how multiplying the 2 outside of the parentheses to the number of unit tiles added for **one** square also gives the total number of unit tiles added.

$$2(x^2 - 8x + \underline{\hspace{1cm}}) - 5$$


- h. Complete the expression to show the subtraction of the total number of unit tiles so the new expression remains equivalent to the original expression, $2x^2 - 16x - 5$.

$$2(x^2 - 8x + \underline{\hspace{1cm}}) - 5 - \underline{\hspace{1cm}}$$

- i. Use the completed squares and the added tiles to rewrite the expression $2x^2 - 16x - 5$.

$$2(x \underline{\hspace{1cm}})^2 \underline{\hspace{1cm}}$$



8.6.3 Bring It Together: Completing the Square Where $a > 1$ Using Algebra Tiles

The diagram of algebra tiles shown represents the expression $4x^2 - 48x - 3$.

	x	-1	-1	-1	-1	-1	-1
x	x^2	$-x$	$-x$	$-x$	$-x$	$-x$	$-x$
-1	$-x$						
-1	$-x$						
-1	$-x$						
-1	$-x$						
-1	$-x$						
-1	$-x$						

	x	-1	-1	-1	-1	-1	-1
x	x^2	$-x$	$-x$	$-x$	$-x$	$-x$	$-x$
-1	$-x$						
-1	$-x$						
-1	$-x$						
-1	$-x$						
-1	$-x$						
-1	$-x$						

	x	-1	-1	-1	-1	-1	-1
x	x^2	$-x$	$-x$	$-x$	$-x$	$-x$	$-x$
-1	$-x$						
-1	$-x$						
-1	$-x$						
-1	$-x$						
-1	$-x$						
-1	$-x$						

	x	-1	-1	-1	-1	-1	-1
x	x^2	$-x$	$-x$	$-x$	$-x$	$-x$	$-x$
-1	$-x$						
-1	$-x$						
-1	$-x$						
-1	$-x$						
-1	$-x$						
-1	$-x$						

-1

-1

-1

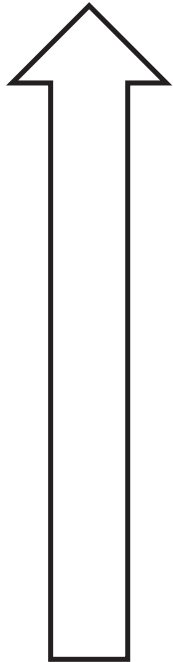
1. Fill in the table to complete the square.

Process	Expression
Original Expression	$4x^2 - 48x - 3$
Factor the first two terms to reflect the algebra tiles in each square. The number factored out is also the number of squares .	$\underline{\hspace{1cm}}(x^2 - \underline{\hspace{1cm}}x) - 3$
Determine the number of unit tiles that should be added for each square.	$\underline{\hspace{1cm}}(x^2 - \underline{\hspace{1cm}}x + \underline{\hspace{1cm}}) - 3$
Multiply the number of squares by the number of unit tiles added to each square. Subtract the total number of unit tiles added.	$\underline{\hspace{1cm}}(x^2 - \underline{\hspace{1cm}}x + \underline{\hspace{1cm}}) - 3 + \underline{\hspace{1cm}}$
Rewrite the perfect square trinomial represented in the square as a binomial squared.	$\underline{\hspace{1cm}}(x - \underline{\hspace{1cm}})^2 - 3 + \underline{\hspace{1cm}}$
Combine like terms.	$\underline{\hspace{1cm}}(x - \underline{\hspace{1cm}})^2 + \underline{\hspace{1cm}}$



8.6.4 Wrap-Up: Reflection

1. Color in the arrow up to the statement which best describes your current understanding.



I'm so confident, I could explain this to someone else!

I can get to the right answer, but I don't understand well enough to explain it yet.

I understand some of this, but I don't understand all of it yet.

I tried hard, and I listened, but I am finding this challenging. I will make sure that I get help.

I do not understand any of this yet. I listed my questions below.

Unit 8, Lesson 7: Completing the Square Algebraically



Benchmarks

MA.912.AR.3.6 Given an expression or equation representing a quadratic function, determine the vertex and zeros, and interpret them in terms of a real-world context.

MA.912.AR.1.2 Rearrange equations or formulas to isolate a quantity of interest.



Learning Targets

- ☐ I can complete the square for a quadratic expression algebraically.
- ☐ I can rewrite a quadratic equation in vertex form by completing the square.